Your Partner Is the Test Suite

Week 11 Thursday — Pair Reasoning: classify a relation under challenge

COMSC 2043 • Discrete Structures and Critical Thinking

October 29, 2026

Today's destination

- Put a classification in writing **before** you talk to anyone.
- Compare representations (pairs, matrix, digraph) and hunt for the **one pair** that separates two positions.
- Challenge another pair's claim with evidence, not volume.
- Say what changed, what survived, and where the model stops being true.

Solo first (10 min): classify this relation

Let $S = \{1, 2, 3, 4\}$ and

$$R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (2,3), (3,2)\}.$$

On your own paper, **before talking**: for each of reflexive, symmetric, antisymmetric, transitive write *yes* or *no*, one line of why, and a confidence from 1 to 5. Then: equivalence, partial order, neither, or both?

Do not look at a neighbor's paper. Wrong answers recorded now are worth more than right answers copied later.

Pairing norms (what is and is not allowed)

Do

- Compare definitions and assumptions first.
- Name the *strongest* part of your partner's reasoning.
- Challenge with a pair, a chain, or a definition.
- Separate "the AI said" from "we checked."

Do not

- Split the work and merge answers.
- Win by repeating louder.
- Assume agreement means correct.
- Copy a partner's write-up. It is yours alone.

Boundary: Checkpoint 2 *replaces* this week's Decision Gate. You submit one, not both.

Warm-up fight: pick a die

Three dice, six faces each. You and your partner each pick one die; higher roll wins.

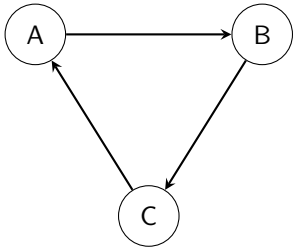
A	2, 2, 4, 4, 9, 9
B	1, 1, 6, 6, 8, 8
C	3, 3, 5, 5, 7, 7

1. Which die would you choose, and why? Argue it with your partner.
2. Define "X beats Y" as: X wins more often than Y. Is this relation **transitive**?

The dice relation, tested

Each matchup has $6 \times 6 = 36$ outcomes:

Matchup	First wins	Chance
A vs B	20 of 36	5/9
B vs C	20 of 36	5/9
C vs A	20 of 36	5/9



No ties; every die has average 5.

So A beats B, B beats C, **and C beats A**.

Classify "beats": reflexive? symmetric? antisymmetric? transitive? Give the witness.

Same relation, three views: what each one makes easy

	Pair list	**Matrix**	**Digraph**
Reflexive	scan for (a, a)	diagonal all 1s	loop on every node
Symmetric	find each reverse	equals its mirror	every arrow has a return
Antisymmetric	hunt two-way pairs	no mirrored off-diagonal 1s	no two-way arrows
Transitive	chain, then look up	hardest to read	every two-step path has a shortcut

Disagree about a property? Switch representations and see whether the **answer stays the same**.

Worked example: the classifier trace

Matrix of R (row a, column b):

	1	2	3	4
1	1	1	0	0
2	1	1	1	0
3	0	1	1	0
4	0	0	0	1

Property		**Evidence**
Reflexive	yes	diagonal all 1s
Symmetric	yes	matrix equals its mirror
Antisymmetric	**no**	$(1,2),(2,1) \in R$, $1 \neq 2$
Transitive	**no**	$(1,2),(2,3) \in R$ but $(1,3) \notin R$

Verdict: neither an equivalence relation nor a partial order. It looks like "is friends with": reflexive and symmetric, but friend-of-a-friend breaks it.

Worked example: repair it, then name the bins

- The two missing pairs are exactly $(1, 3)$ and $(3, 1)$. Add them: that is the **smallest** transitive fix.
- New relation R': reflexive ✓, symmetric ✓, transitive ✓, antisymmetric no.
- So R' is an **equivalence relation** with two bins: $\{1, 2, 3\}$ and $\{4\}$.

Challenge move for your pair

Take your own relation, break exactly one property with a single pair, then find the *smallest* set of pairs that repairs it.

Optional: a classifier in a dozen lines

```
def props(R, S):
    refl  = all((a, a) in R for a in S)
    sym   = all((b, a) in R for (a, b) in R)
    anti  = all(a == b for (a, b) in R if (b, a) in R)
    trans = all((a, d) in R
                for (a, b) in R for (c, d) in R if b == c)
    return refl, sym, anti, trans
```

Each line is one definition. If your code disagrees with your hand trace, **one of them is wrong**, and finding which is the lesson.

The discriminating witness

Challenge script

"Your claim says _____. Check the pair or chain _____."

- A **yes** needs a check of every pair, or a proof.
- A **no** needs exactly one witness.
- If you cannot find a witness, say so, and then say what you checked.

Strong challenge: points at a pair. **Weak challenge:** "I just don't think so."

Puzzle: the prerequisite deadlock

The registrar's rule: "X must be taken before Y."

1. Algorithms before Data Structures before Systems before **Algorithms**. How many valid orders of the three courses exist?
2. Same three courses, but only Algorithms $\to$ Data Structures $\to$ Systems. How many valid orders?
3. Four courses in a diamond: $A \to B$, $A \to C$, $B \to D$, $C \to D$. How many?

Which of the four properties does the cycle break, and why does a partial order forbid it?

Puzzle: how many relations?

On a 3-element set there are $2^9 = 512$ relations. Predict, then compare with your partner:

1. How many are **reflexive**?
2. How many are **symmetric**?
3. How many are **both** reflexive and symmetric?
4. How many are equivalence relations? Partial orders?

Hint: for each one, count the pairs you are *free* to choose.

Counts, verified by brute force

On 3 elements	Count
All relations	512
Reflexive	64
Symmetric	64
Reflexive and symmetric	8
Equivalence relations	5
Partial orders	19

Most relations are neither. Structure is rare, and the four properties are how you find it.

Cross-pair challenge (13 min)

1. Pair up with another pair. Each pair names **one** claimed property of its relation.
2. The other pair gets 4 minutes to break it: a pair, a chain, or a definition.
3. Reply: *concede*, *narrow the claim*, or *show the check that holds*.
4. Swap roles.

Record the **evidence that mattered**, not who won.

Individual write-up (14 min)

Your own words, one or two sentences per field:

1. **Starting position**: what I believed before talking.
2. **Strongest challenge**: what my partner raised.
3. **Evidence that mattered**: the pair, chain, or definition.
4. **What changed or survived**.
5. **Current position** and why.
6. **What would change my mind next**.

Then use the remaining time on **Checkpoint 2**: revise, test, or delimit one earlier claim.

Where the model stops being true

A relation is only as good as its **domain** and its **definition**.

- "Beats" for the dice means *wins more often*. Change it to "higher average" and it becomes transitive, but it is a different relation.
- "Is a sibling of": does it include half-siblings? The property you find depends on the answer.
- Divisibility is a partial order on the positive integers; on all integers it is not antisymmetric ($2 \mid -2$ and $-2 \mid 2$).

Write one boundary sentence: "My classification holds for _____ and would change if _____."

Preview: when the relation becomes a network

- A digraph *is* a relation drawn as arrows.
- Next week: paths, connectivity, traversal, shortest path.
- The question: **are we solving the graph correctly, or modeling reality incorrectly?**

Bring one thing you could model as nodes and arrows, and say what an **arrow means**.

